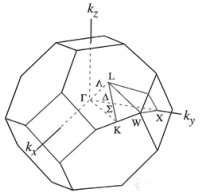
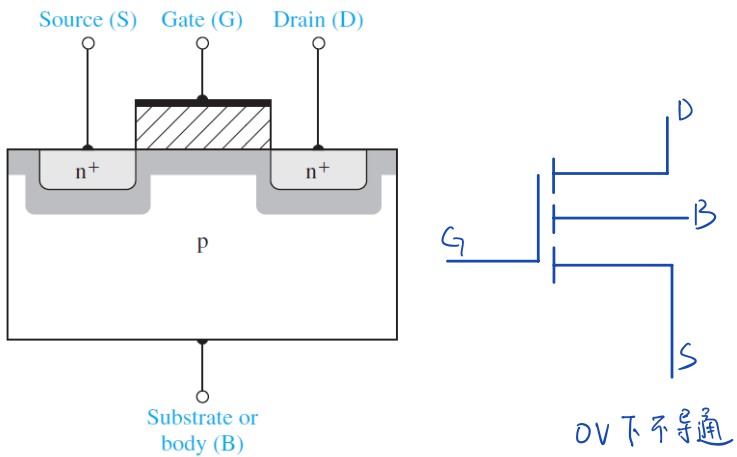
Basic MOSFET Operation



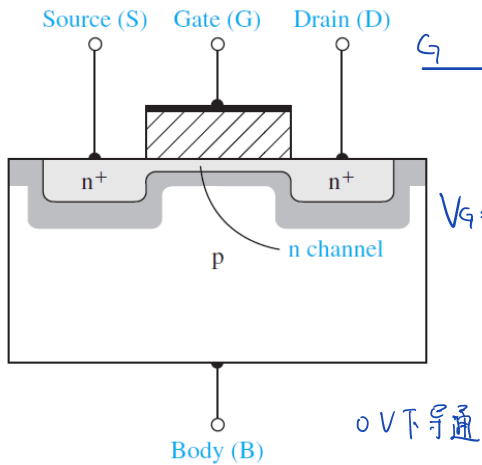
MOSFET

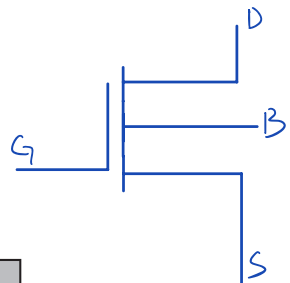
- Metal-Oxide-Semiconductor Field Effect Transistor
- Enhancement mode
  - ➢The semiconductor under oxide is not inverted with zero gate voltage
- Depletion mode
  - ➢The semiconductor under oxide is inverted with zero gate voltage



Enhancement mode

0V下不导通

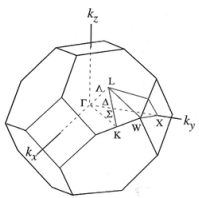




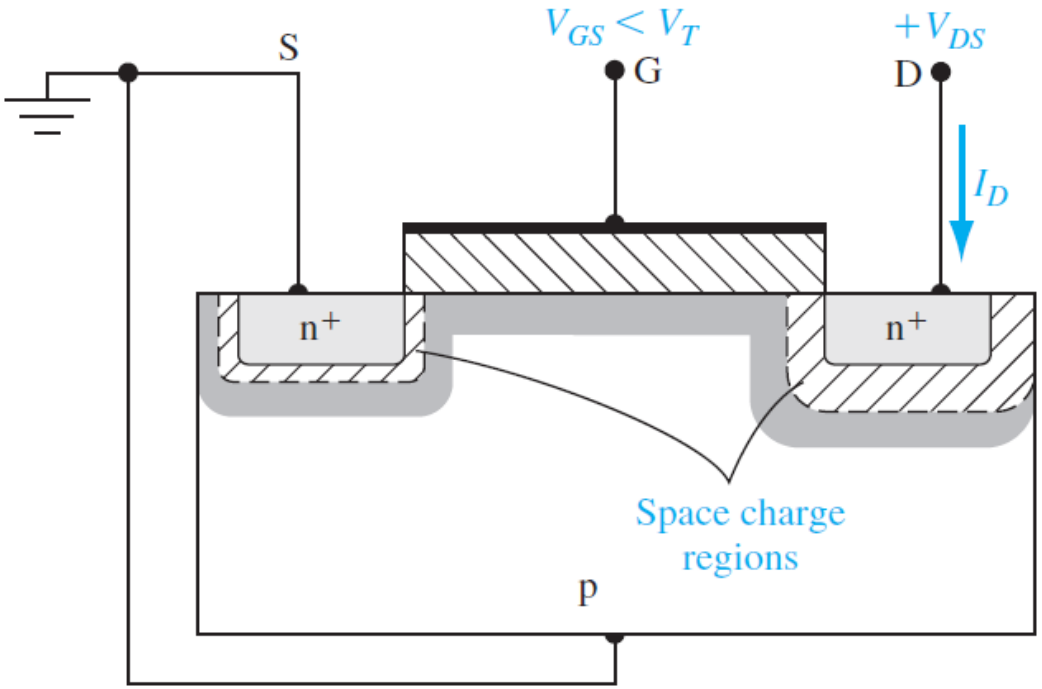
$V_G = 0$. 沟道中有反型载流子

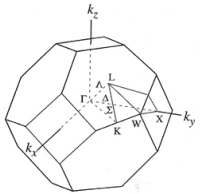
0V下导通

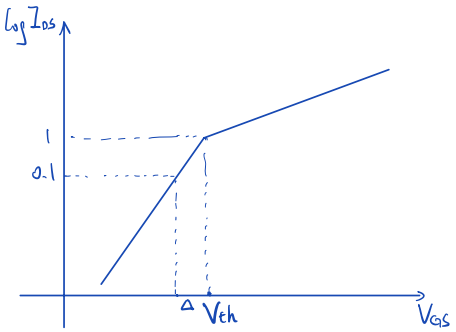
Depletion mode



I-V Relationship $V_{gs}<V_{th}$



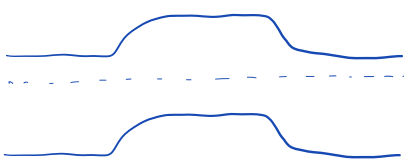




$I_{DS} \rightarrow 0.1$

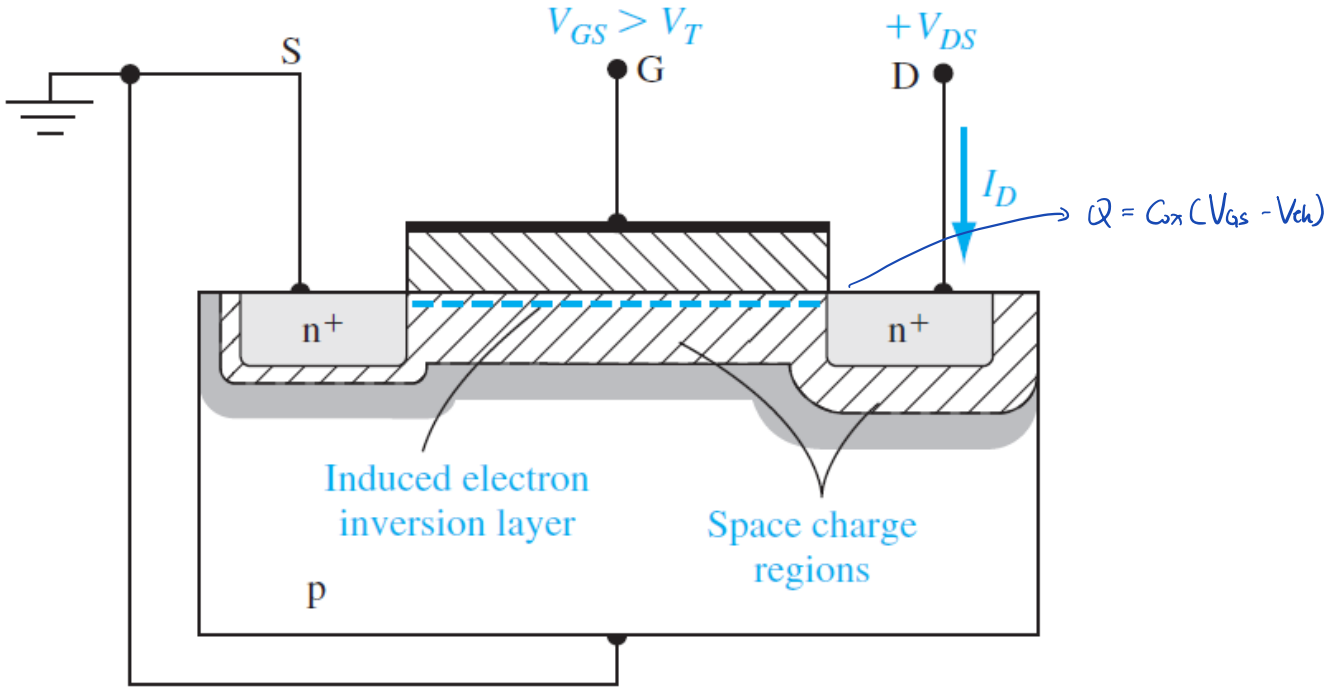
$V_{GS} \rightarrow \underbrace{\Delta}$

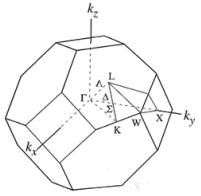
Sub threshold swing



$$I \propto \frac{e^{\frac{e\Delta V_{GS}}{KT}}(}{L}$$

I-V Relationship $V_{gs}>V_{th}$

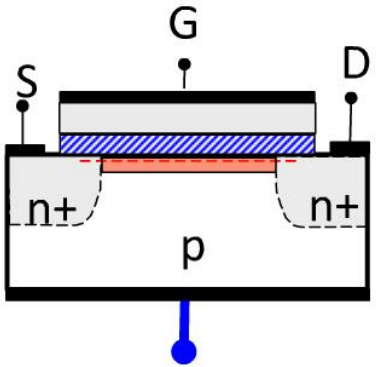


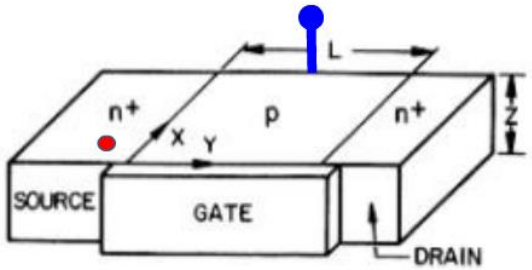


Small V_{DS}

$$I_D = g_d V_{DS}$$

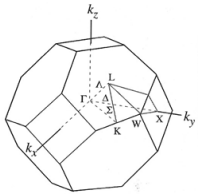
$$g_d = \frac{W}{L} \cdot \mu_n |Q_n'|$$





$|Q_n'|$ is the inversion layer charge per unit area

$$|Q_n'| = C_{ox}(V_{GS} - V_{th}) \Rightarrow I_D = \frac{W}{L} \mu_n C_{ox} \cdot V_{DS} (V_{GS} - V_{th})$$

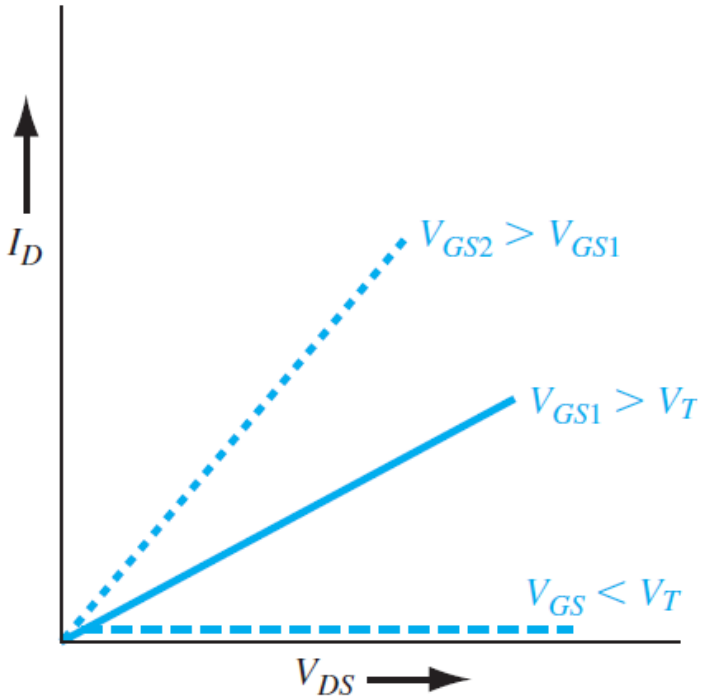


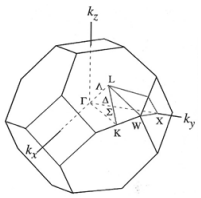
I_D v.s V_{DS} for Small V_{DS}

$$I_D = g_d V_{DS}$$

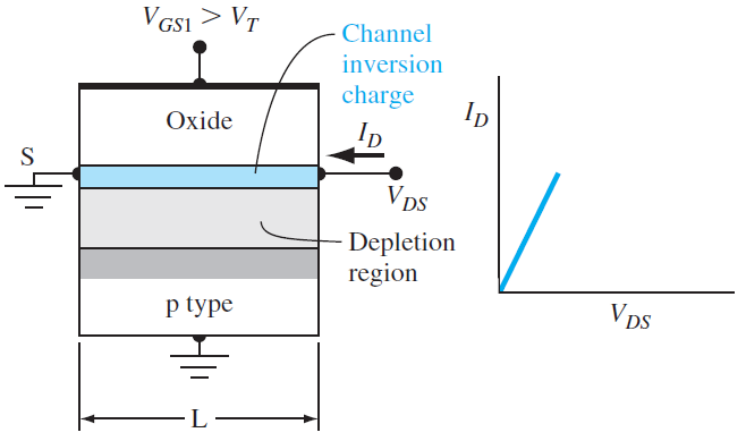
$$g_d = \frac{W}{L} \cdot \mu_n |Q_n'|$$

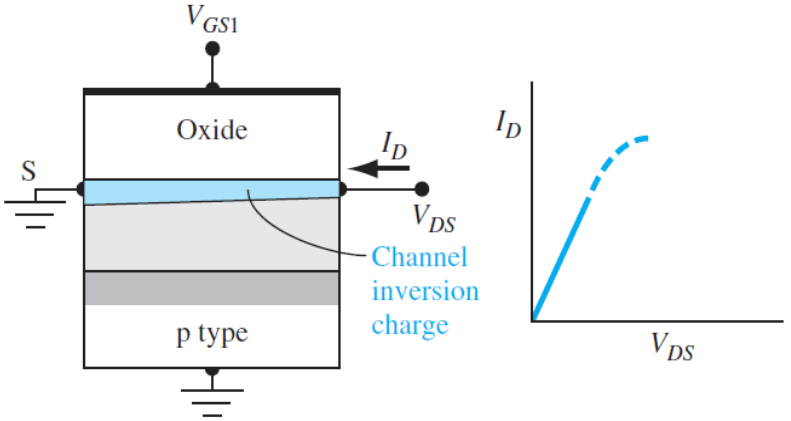
$$|Q_n'| = C_{ox}(V_{GS} - V_{th})$$

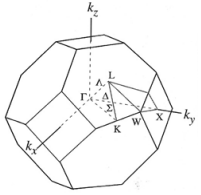




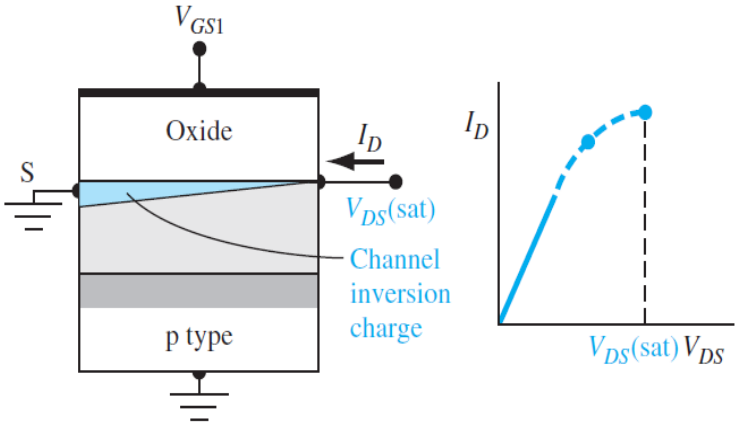
As V_{DS} Becomes Larger

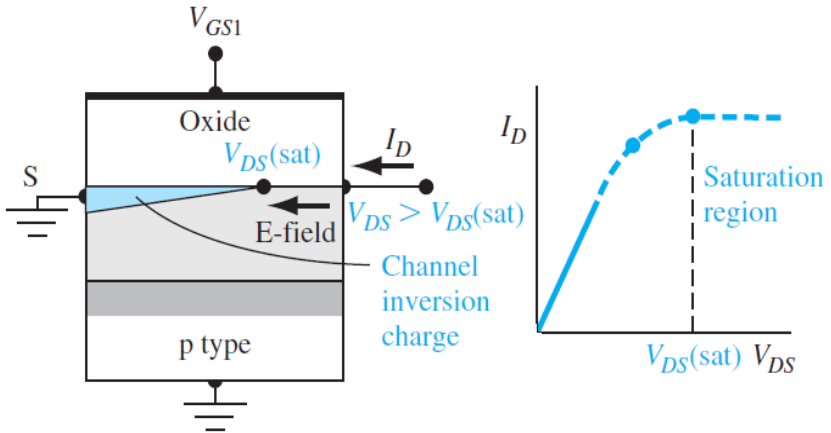






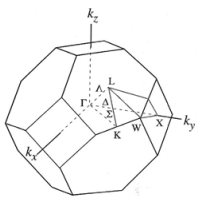
As V_{DS} Becomes Larger

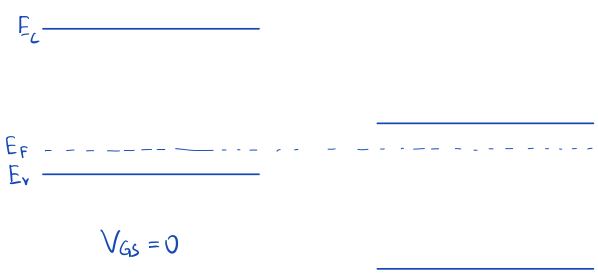




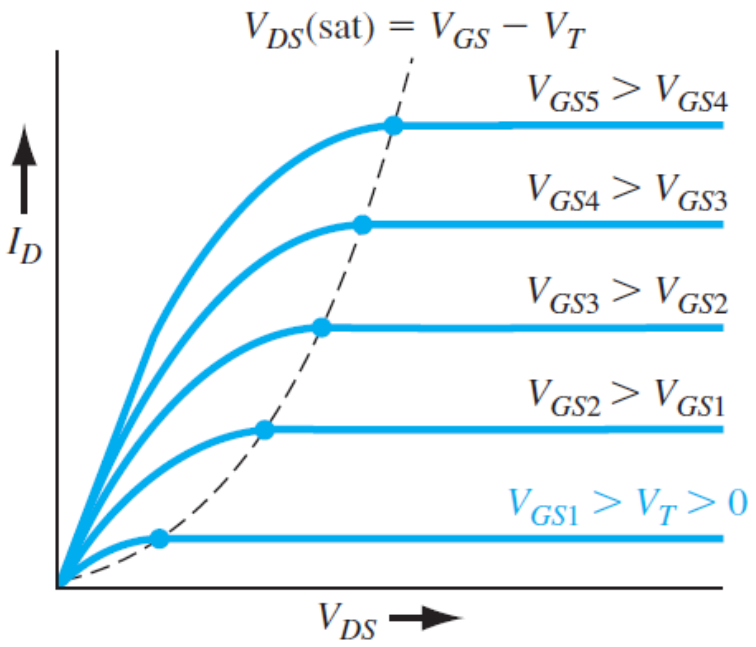
$$V_{DS}(\text{sat}) = V_{GS} - V_T$$

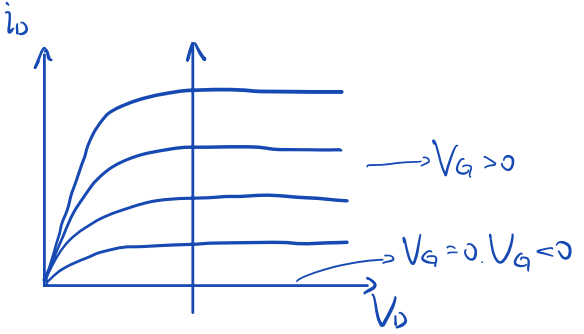
$V_D = 1.1$ V

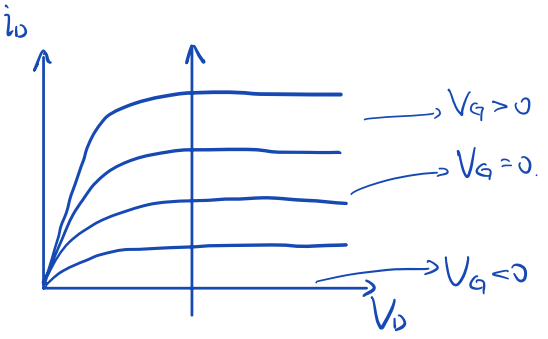


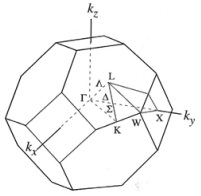

E_c
E_F
E_v
$V_{GS} = 0$

Family of I_D v.s V_{DS}









I-V Relationship Mathematical Derivation

$$I_x = \int_y \int_z J_x \, dy \, dz$$

$$I_x = -W\mu_n Q_n' \mathrm{E}_x$$

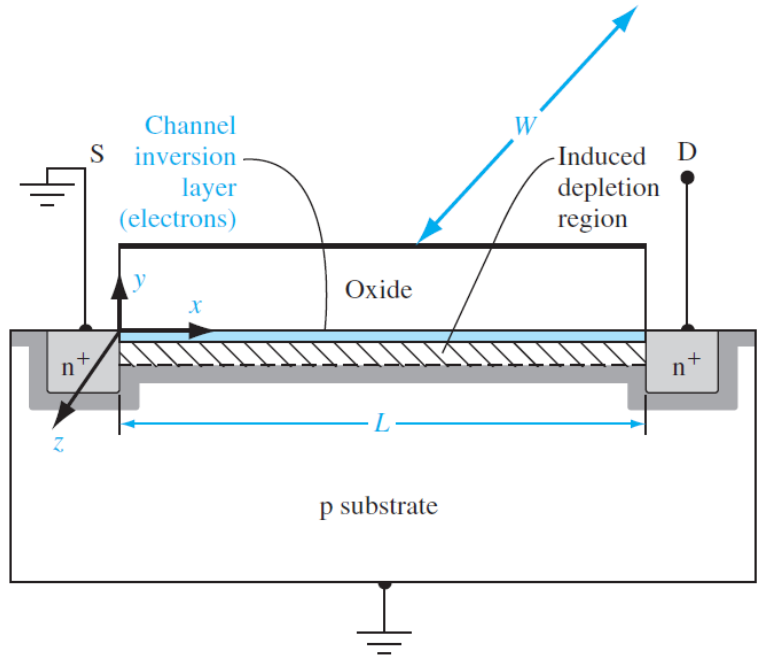
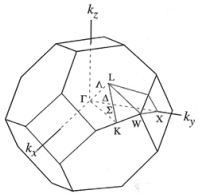


Figure 10.43 | Geometry of a MOSFET for I_D versus V_{DS} derivation.

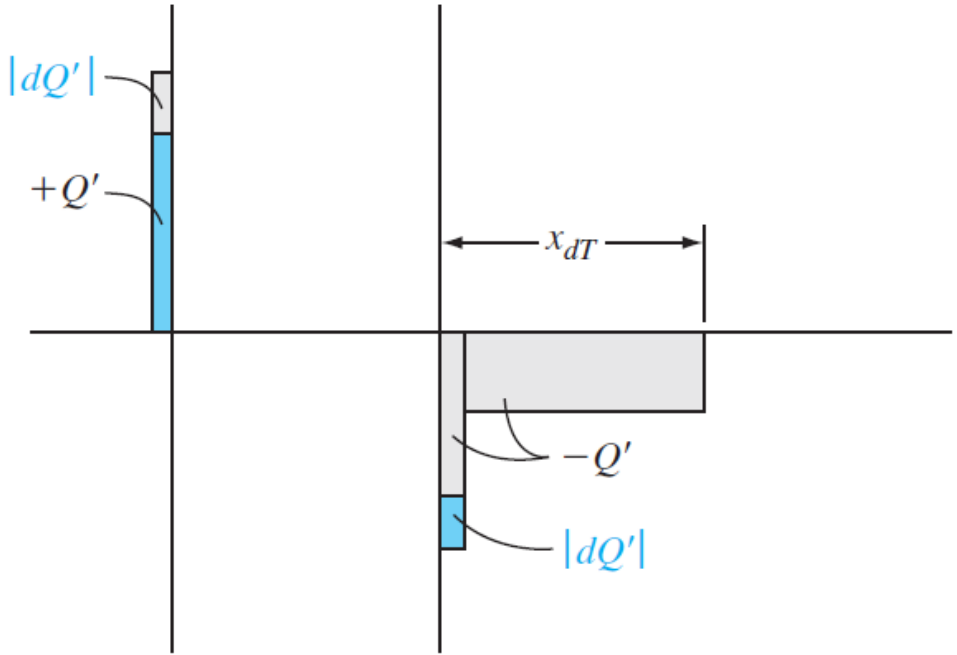
$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$

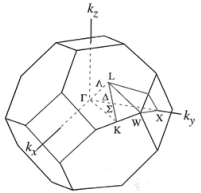
短沟道器件 Gate 对沟道的控制能力↓
其它结点对沟道的影响↑



I-V Relationship Mathematical Derivation

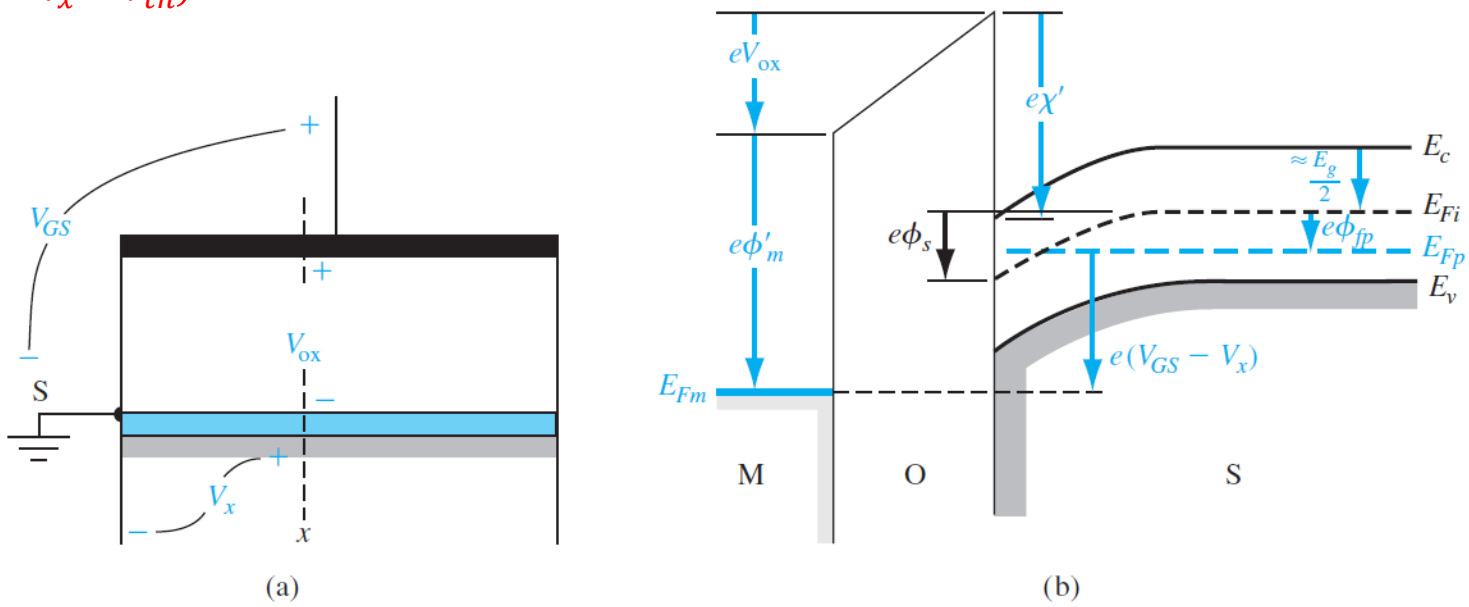
What is the $|Q_n'|$?





I-V Relationship Mathematical Derivation

$$|Q_n'| = C_{ox}(V_{GS} - V_x - V_{th})$$



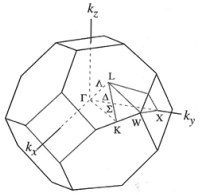


Figure 10.46 | (a) Potentials at a point x along the channel. (b) Energy-band diagram through the MOS structure at the point x.

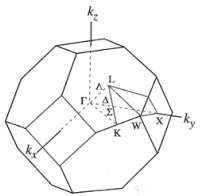
I-V Relationship Mathematical Derivation

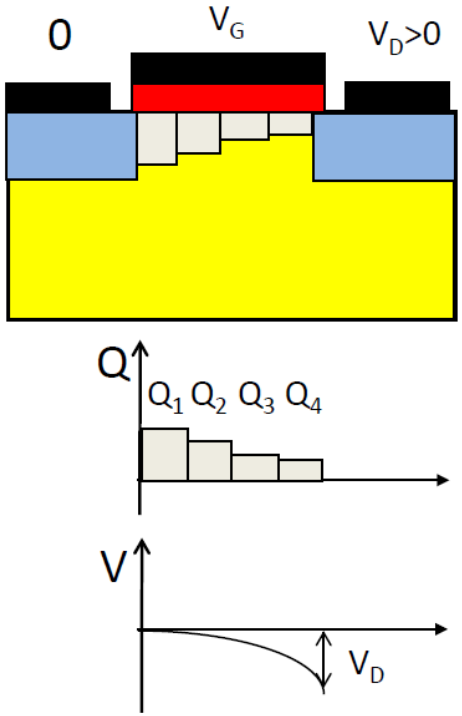
$$|Q_n'| = C_{ox}(V_{GS} - V_x - V_{th})$$

$$I_x = -W\mu_n Q_n' \mathrm{E}_x$$

$$I_x = -W\mu_n C_{\mathrm{ox}} \frac{dV_x}{dx}[(V_{GS} - V_x) - V_T]$$

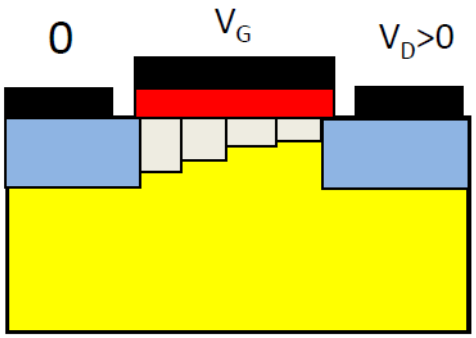
$$\int_0^L I_x\, dx = -W\mu_n C_{\mathrm{ox}} \int_{Vx(0)}^{Vx(L)} [(V_{GS} - V_T) - V_x]\, dV_x$$

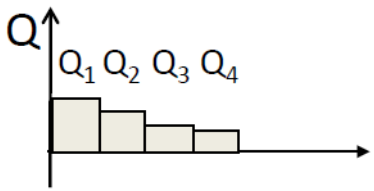


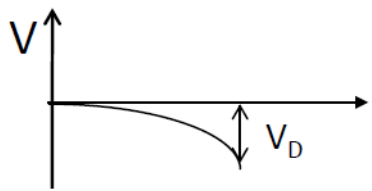


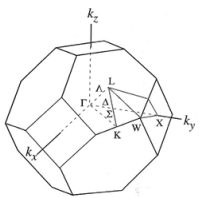
I-V Relationship Mathematical Derivation

$$I_D = \frac{W\mu_n C_{ox}}{2L}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$









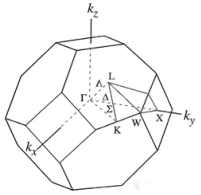
I-V Relationship Mathematical Derivation

$$I_D = \frac{W\mu_n C_{ox}}{2L}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

$$V_{DS} > V_{DS}(\text{sat})$$

$$I_D(\text{sat}) = \frac{W\mu_n C_{ox}}{2L}[2(V_{GS} - V_T)V_{DS}(\text{sat}) - V_{DS}^2(\text{sat})]$$

$$I_D(\text{sat}) = \frac{W\mu_n C_{ox}}{2L}(V_{GS} - V_T)^2$$



In modern transistor, we never find this relationship experimentally

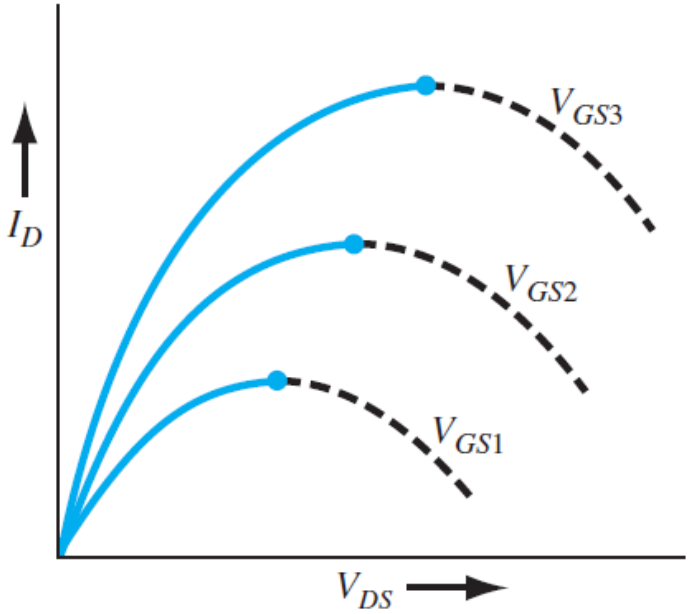


Figure 10.47 | Plots of I_D versus V_{DS} from Equation (10.62).

Transconductance

- Transconductance: the change in drain current with respect to the corresponding change in gate voltage.

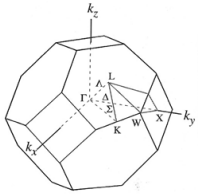
$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

$$I_D = \frac{W\mu_n C_{ox}}{2L}\left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right]$$

$$g_{mL} = \frac{\partial I_D}{\partial V_{GS}} = \frac{W\mu_n C_{ox}}{L} \cdot V_{DS}$$

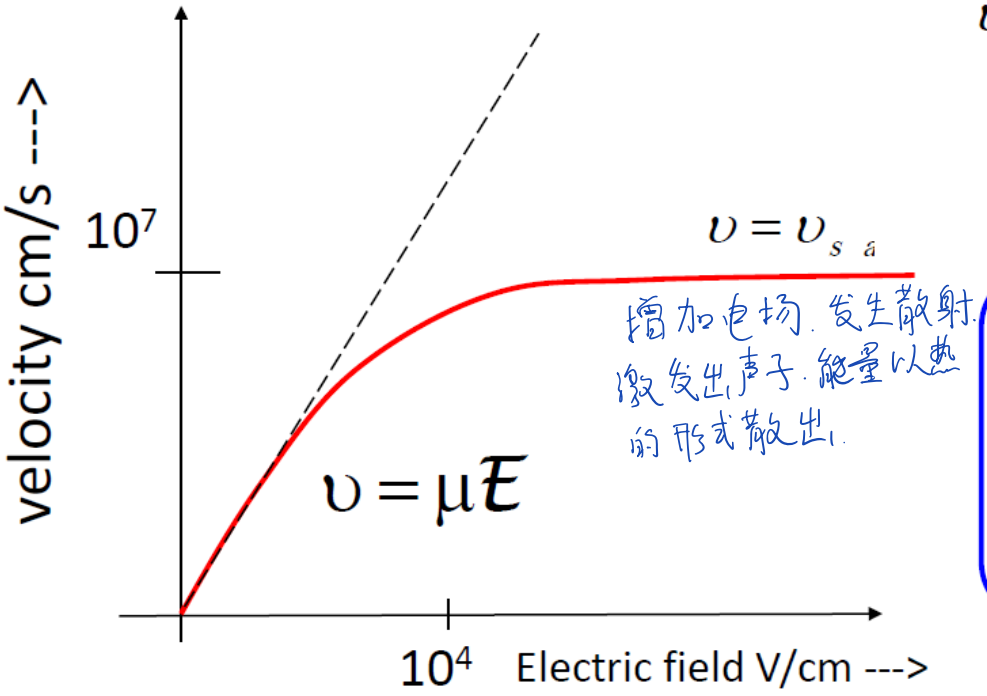
$$I_D\,(\text{sat}) = \frac{W\mu_n C_{ox}}{2L}(V_{GS} - V_T)^2$$

$$g_{ms} = \frac{\partial I_D(\text{sat})}{\partial V_{GS}} = \frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)$$



pinch off

Velocity vs. Field Characteristic (electrons)

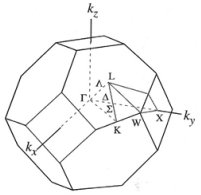


增加电场，发生散射，激发出声子，能量以热的形式散出。

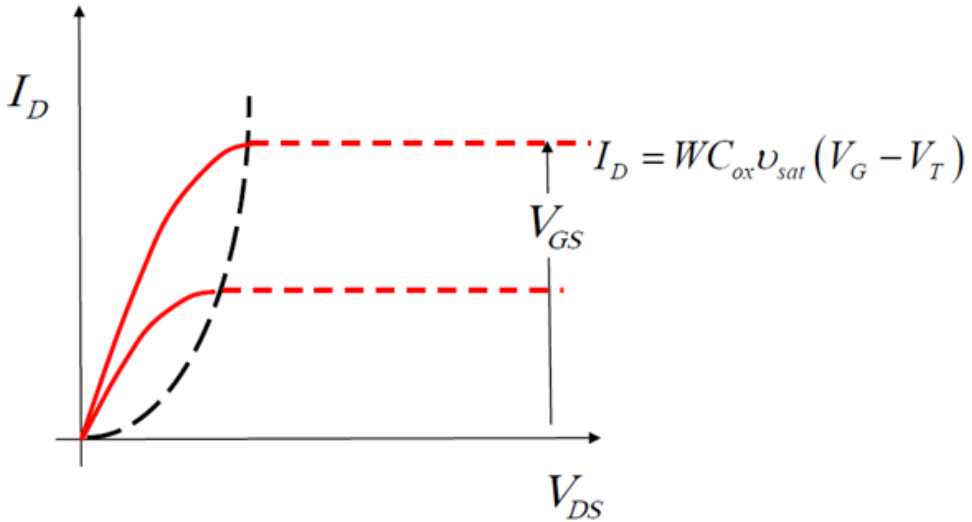
$$\upsilon_d = \frac{-\mu\mathcal{E}}{\left[1+(\mathcal{E}/\mathcal{E}_c)^2\right]^{1/2}}$$

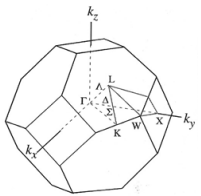
$$\upsilon_d = \frac{-\mu\mathcal{E}}{1+(|\mathcal{E}|/\mathcal{E}_c)}$$

$$\upsilon_{d,sat} = \mu\mathcal{E}_c$$



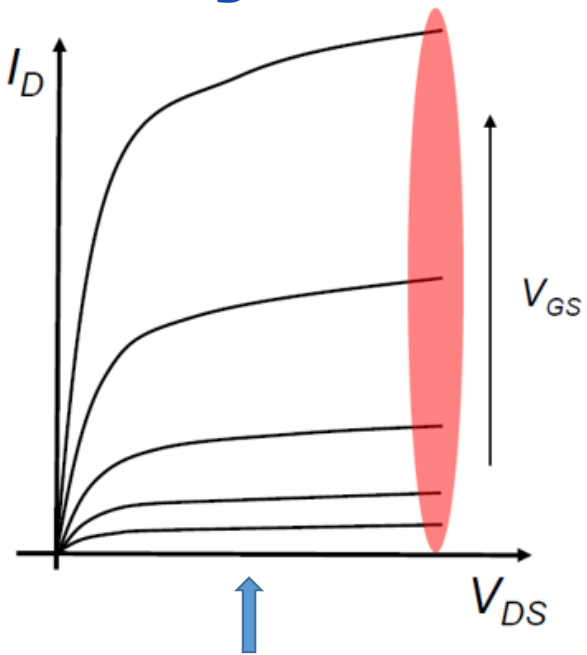
Velocity Saturation



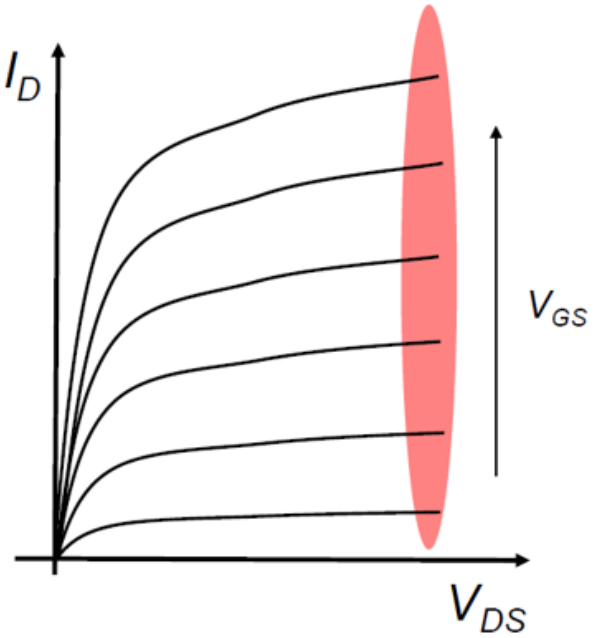


'Signature' of Velocity Saturation

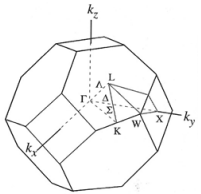
long channel transistor



short channel transistor.



We never see this practically for more than 20 years!



Substrate Bias Effects

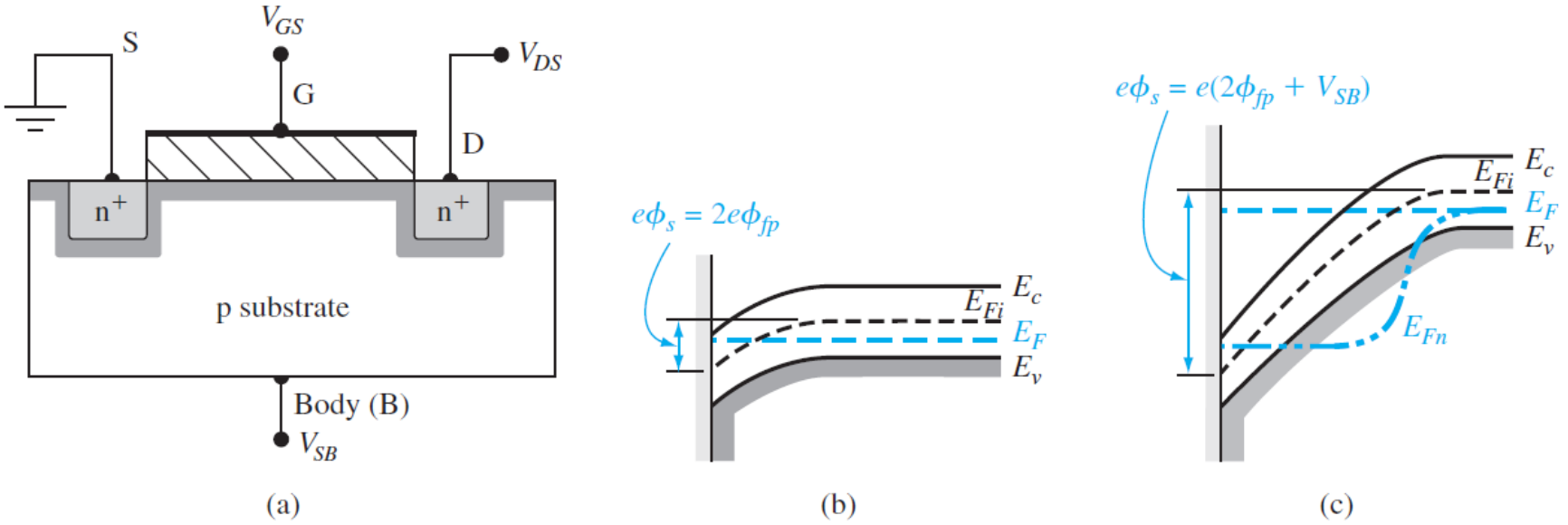
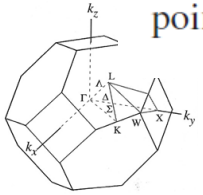


Figure 10.50 | (a) Applied voltages on an n-channel MOSFET. (b) Energy-band diagram at inversion point when $V_{SB} = 0$. (c) Energy-band diagram at inversion point when $V_{SB} > 0$ is applied.



Substrate Bias Effects

$V_{SB} = 0$

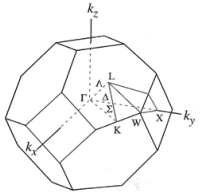
$$Q'_{SD}\,(\max) = -eN_a x_{dT} = -\sqrt{2e\epsilon_s N_a(2\phi_{fp})}$$

$V_{SB} > 0$

$$Q'_{SD} = -eN_a x_d = -\sqrt{2e\epsilon_s N_a(2\phi_{fp} + V_{SB})}$$

$$\Delta Q'_{SD} = -\sqrt{2e\epsilon_s N_a}\left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}\right]$$

$$\Delta V_T = -\frac{\Delta Q'_{SD}}{C_{ox}} = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}\left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}\right]$$



Frequency Limitations

- Small-Signal Equivalent Circuit

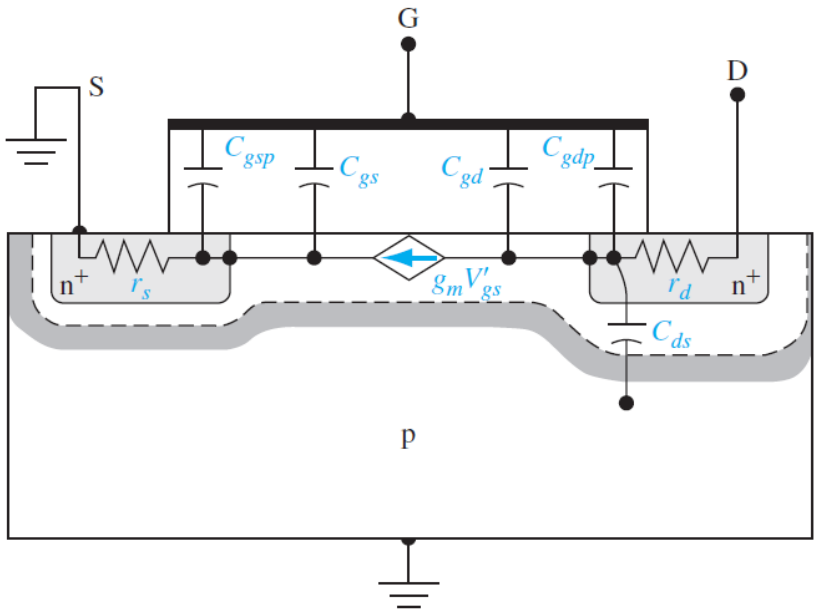
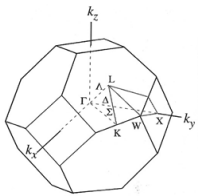


Figure 10.52 | Inherent resistances and capacitances in the n-channel MOSFET structure.



Small-Signal Equivalent Circuit

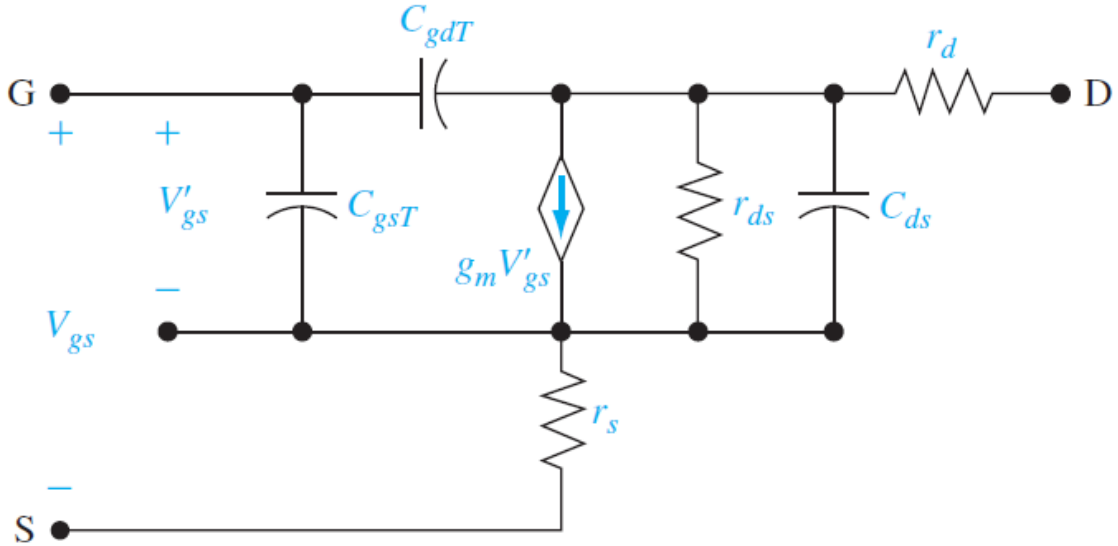
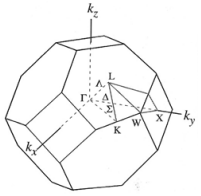


Figure 10.53 | Small-signal equivalent circuit of a common-source n-channel MOSFET.



Impact from Source Resistance r_s

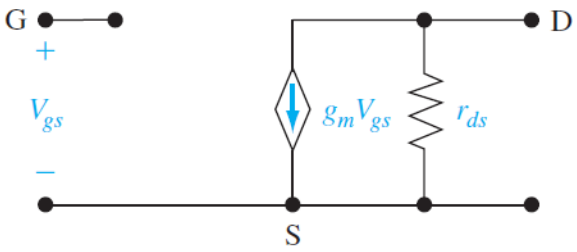


Figure 10.54 | Simplified, low-frequency small-signal equivalent circuit of a common-source n-channel MOSFET.

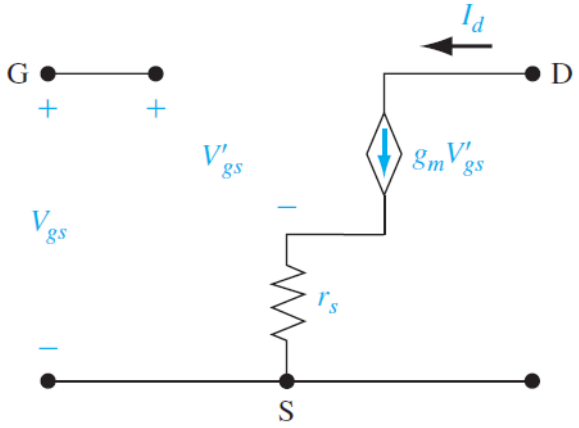
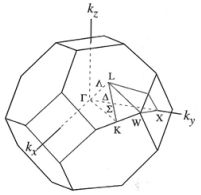


Figure 10.55 | Simplified, low-frequency small-signal equivalent circuit of common-source n-channel MOSFET including source resistance r_s.

$$I_d = g_m V'_{gs}$$

$$V_{gs} = V'_{gs} + (g_m V'_{gs}) r_s = (1 + g_m r_s) V'_{gs}$$

$$I_d = \left(\frac{g_m}{1 + g_m r_s} \right) V_{gs} = g'_m V_{gs}$$



Frequency Limitation Factors

- Channel transit time
  - Transit time: $\tau_t = L/v_{\text{sat}}$
  - $v_{sat} = 10^7$ cm/s, L=1μm
  - Transit time limit frequency >100GHz
- Capacitance charging time

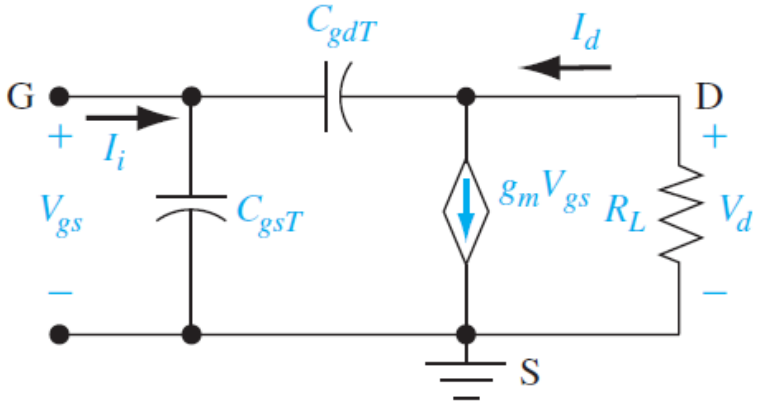
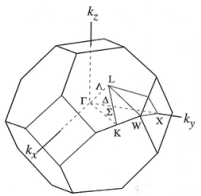


Figure 10.56 | High-frequency small-signal equivalent circuit of common-source n-channel MOSFET.



Frequency Limitation Factors

$$I_i = j\omega C_{gsT} V_{gs} + j\omega C_{gdT}(V_{gs} - V_d)$$

$$\frac{V_d}{R_L} + g_m V_{gs} + j\omega C_{gdT}(V_d - V_{gs}) = 0$$

$$I_i = j\omega\left[C_{gsT} + C_{gdT}\left(\frac{1 + g_m R_L}{1 + j\omega R_L C_{gdT}}\right)\right]V_{gs}$$

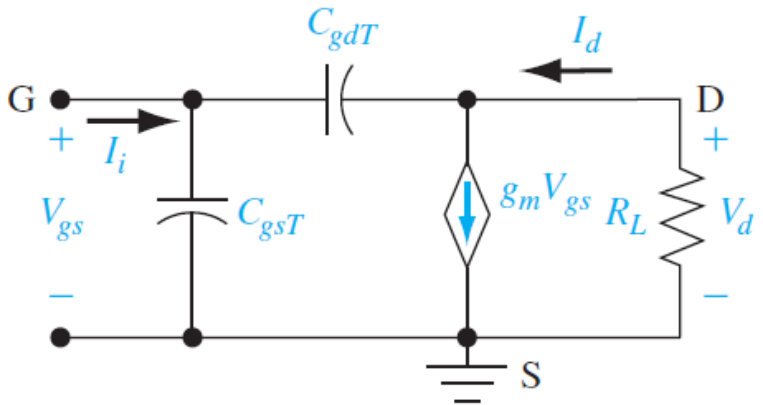
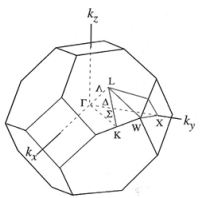


Figure 10.56 | High-frequency small-signal equivalent circuit of common-source n-channel MOSFET.



Miller Capacitance

$$I_i = j\omega\left[C_{gsT} + C_{gdT}\left(\frac{1 + g_m R_L}{1 + j\omega\, R_L C_{gdT}}\right)\right]V_{gs}$$

$$I_i = j\omega[C_{gsT} + C_{gdT}(1 + g_m R_L)]V_{gs}$$

$$C_M = C_{gdT}(1 + g_m R_L)$$

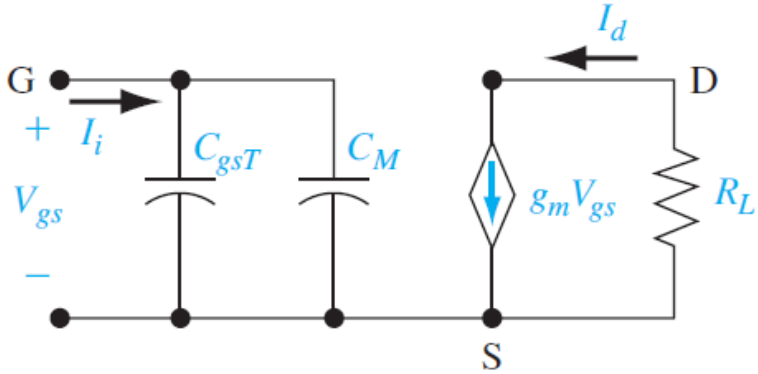
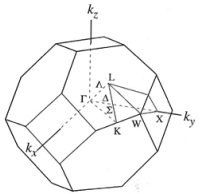


Figure 10.57 | Small-signal equivalent circuit including Miller capacitance.



Miller Capacitance

$$I_i = j\omega[C_{gsT} + C_{gdT}(1 + g_m R_L)]V_{gs}$$

$$C_M = C_{gdT}(1 + g_m R_L)$$

$$I_i = j\omega(C_{gsT} + C_M)V_{gs}$$

$$I_d = g_m V_{gs}$$

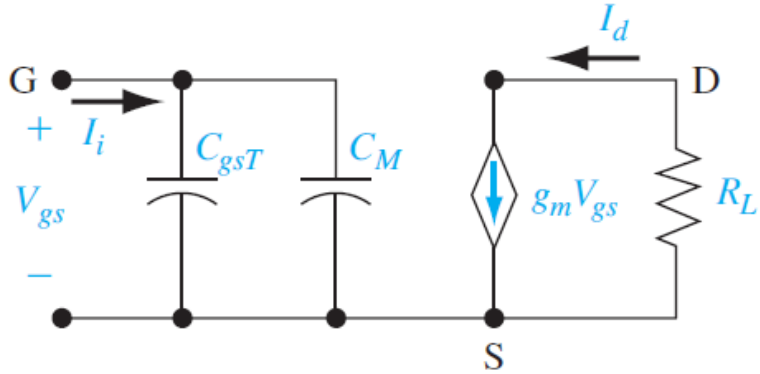
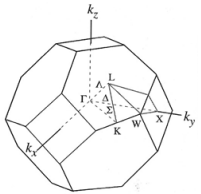


Figure 10.57 | Small-signal equivalent circuit including Miller capacitance.



Cutoff Frequency

$$\left|\frac{I_d}{I_i}\right| = \frac{g_m}{2\pi f(C_{gsT} + C_M)}$$

$$f_T = \frac{g_m}{2\pi(C_{gsT} + C_M)} = \frac{g_m}{2\pi C_G}$$

Cutoff frequency f_T = frequency where MOSFET no longer amplifies input (gate) signal

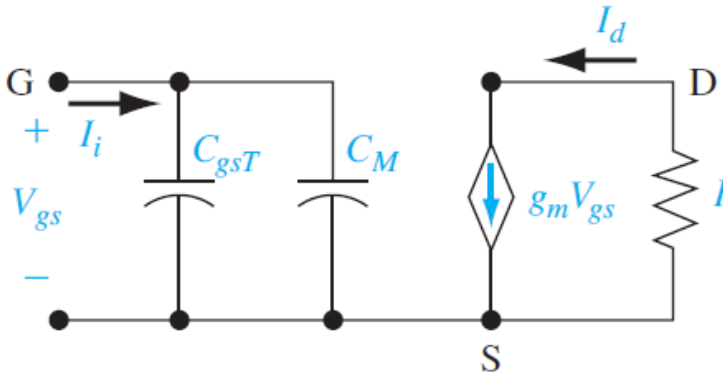
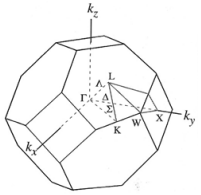


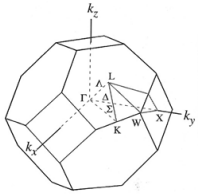
Figure 10.57 | Small-signal equivalent circuit including Miller capacitance.



Ideal Case

$$g_{ms} = \frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T) \qquad f_T = \frac{g_m}{2\pi C_G} = \frac{\frac{W\mu_n C_{ox}}{L}(V_{GS} - V_T)}{2\pi(C_{ox}WL)} = \frac{\mu_n(V_{GS} - V_T)}{2\pi L^2}$$

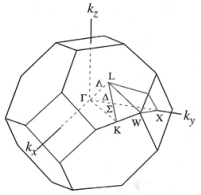
- Something we already knew qualitatively → higher MOSFET operating frequency
  - ✓ achieved by decreasing channel
  - ✓ length L, increasing mobility μ_{eff}
- Smaller = faster for devices (though parasitic capacitances play a big role in realistic circuits)



Example

Objective: Calculate the cutoff frequency of an ideal MOSFET with a constant mobility.

Assume that the electron mobility in an n-channel device is $\mu_n = 400$ cm^2/V-s and that the channel length is $L = 4$ μm. Also assume that $V_T = 1$ V and $V_{GS} = 3$ V.



Example

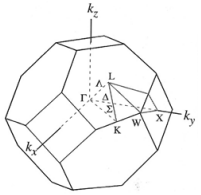
Objective: Calculate the cutoff frequency of an ideal MOSFET with a constant mobility.

Assume that the electron mobility in an n-channel device is $\mu_n = 400$ cm^2/V-s and that the channel length is $L = 4$ μm. Also assume that $V_T = 1$ V and $V_{GS} = 3$ V.

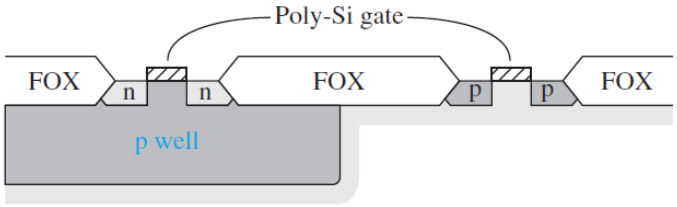
■ Solution

From Equation (10.96), the cutoff frequency is

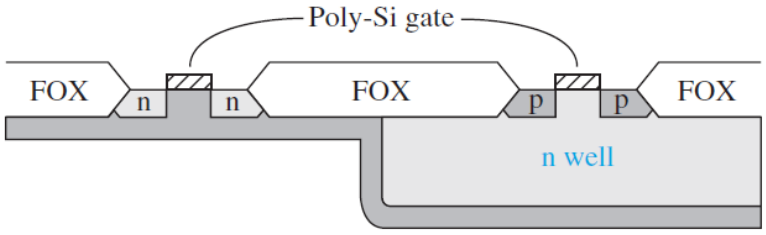
$$f_T = \frac{\mu_n(V_{GS} - V_T)}{2\pi L^2} = \frac{400(3-1)}{2\pi\,(4 \times 10^{-4})^2} = 796 \text{ MHz}$$



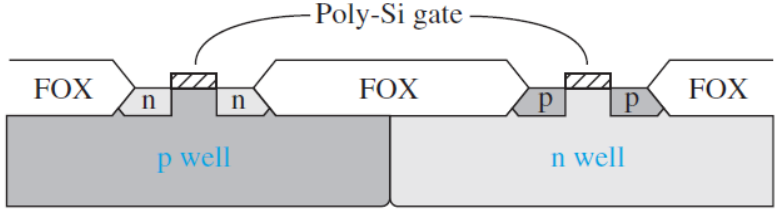
Complementary MOS (CMOS) Technology



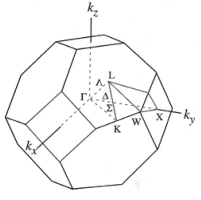
n substrate



p substrate

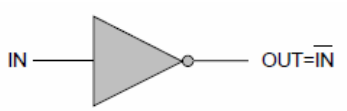


p or n substrate

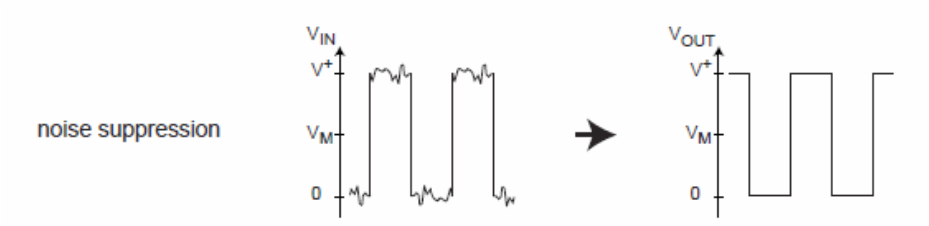


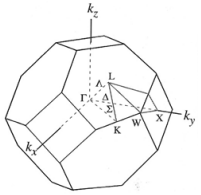
Logic Application

- CMOS Inverter Circuit
- Key property: signal regeneration – returns logic outputs (0 or $1=V^{+}=V_{DD}$) even in presence of noise



IN	OUT
0	1
1	0

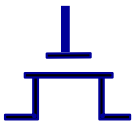




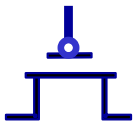
CMOS Inverter Circuit

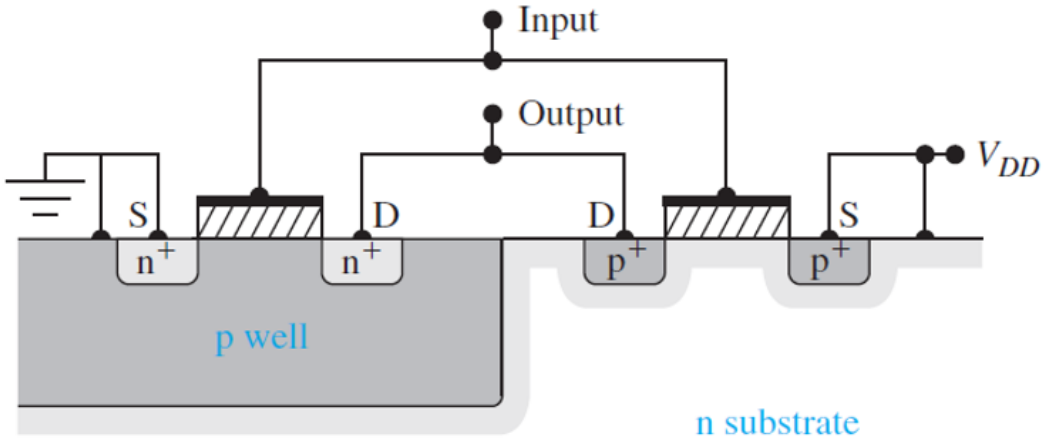
CIRCUIT SYMBOLS

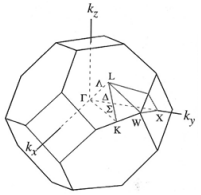
N-channel MOSFET



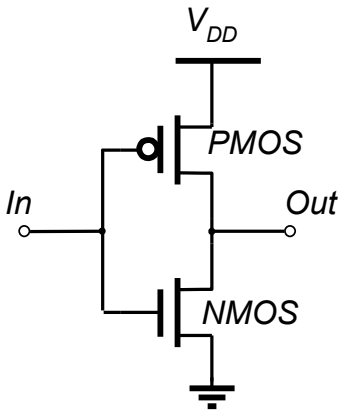
P-channel MOSFET

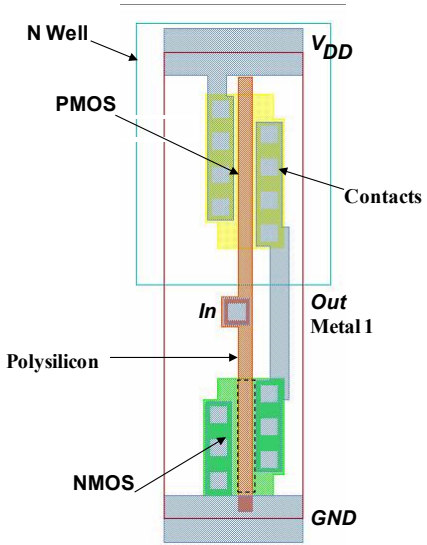


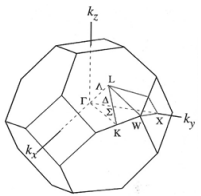




CMOS Inverter Circuit

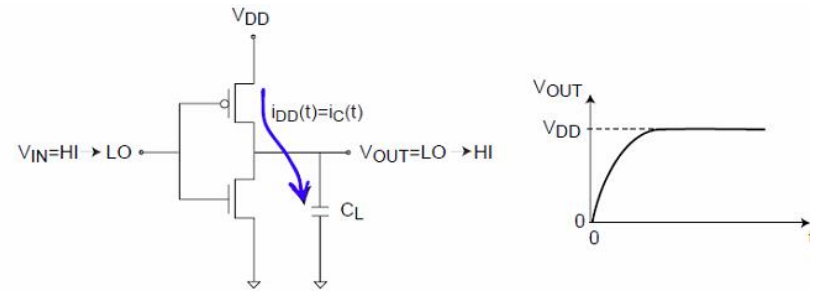




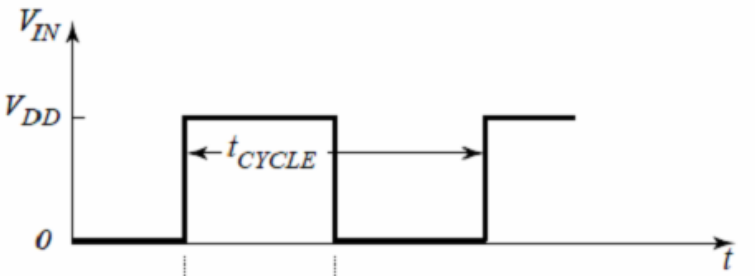


A Quick Look at CMOS Power Dissipation

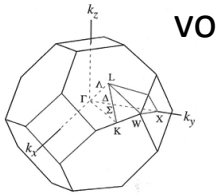
- Energy consumed while charging capacitive load: E_P



- C_L is discharged through NFET → E_N



- Total energy dissipated per clock cycle: $E = C_L V_{DD}^2$
- Frequency *f* cycles per second → active power $P = fC_L V_{DD}^2$
- This is very important: <u>fundamental trade-off between speed (f) and power dissipation</u>. Reducing voltage and parasitic C's is a must to keep power low at higher speeds.



Leakage Power

- In reality, there is also passive power (leakage) dissipated by the FETs supposed to be "off": $P_{off} = I_{leak}V_{DD}$
- $I_{off} \sim I_{on}/1000$ in modern technology per transistor
- But this can become a headache when you have 100s of millions of "sleeping" transistors (i.e. "passive power" vs. "active power")!

